[Metric]
MIL-DTL-18455D
17 August 2021
SUPERSEDING
M1L-A-18455C
23 December 1986

DETAIL SPECIFICATION
ARGON, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of argon gas for use in inert gas shielded electric arc welding of aluminum, magnesium, corrosion-resisting steel, and other alloys.

1.2 Classification. The types of argon covered by this specification are as follows (see 6.2):

Type I - Gaseous
Type II - Liquid

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFCATIONS

RR-C- 901	-	Cylinders, Compressed Gas: Seamless Shatterproof, High Pressure DOT 3AA Steel, and 3AL Aluminum

Comments, suggestions, or questions on this document should be addressed to the DLA Aviation, VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil

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FEDERAL STANDARDS

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-313	-	Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities

COMMERCIAL ITEM DESCRIPTIONS

A-A-59860	-	Valves, Cylinder, Gas (for Compressed or Liquefied Gases)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-27626	-	Sampler, Cryogenic Liquid

DEPARTMENT OF DEFENSE STANDARDS

MIL-T-704	-	Treatment and Painting of Materials
MIL-STD-101	-	Color Code for Pipelines and for Compressed Gas Cylinders
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-1411	-	Inspection and Maintenance of Compressed Gas Cylinders
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

Title 29, Part 1910.1200	-	Hazard Communication
Title 49	-	Transportation

(Copies of these documents are available online at https://www.gpo.gov/fdsys/browse/collectionCfr.action?collectionCode=CFR.)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

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Research paper No. 1865 -	Measurement of Water in Gases by Electrical Conduction in a Film of Hygroscopic Material and the Use of Pressure Changes in Calibration.
Research paper No. 1381 -	Apparatus for Absorption of Gravimetric Determination of Constituents for a Gas Mixture.

(Copies of these documents are available online at https://www.nist.gov/nist-research-library/journal-research-nist.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN CHEMICAL SOCIETY (ACS)

Analytical Chemistry Volume 26, No. 3 -	"Determination of Nitrogen Content of Noble Gases - Absorption in Titanium Metal", by H.S. Dombrowski

(Copies of these documents are available online at https://pubs.acs.org/.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ H1331	-	Zero Acceptance Number Sampling Plans

(Copies of this document are available from https://www.asq.org/.)

COMPRESSED GAS ASSOCIATION (CGA)

G-11.1	-	Commodity Specification for Argon

(Copies of these documents are available online at https://www.cganet.com/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Composition. The argon shall not contain oil or odorous or toxic impurities, shall be as specified in Table I, and shall be tested in accordance with 4.4.1-4.4.4.

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TABLE I. Composition of Argon.

Oxygen, max.	50 ppm
Hydrogen, max.	50 ppm
Nitrogen, max.	50 ppm
Argon, min.	99.985 %

3.2 Moisture content. When tested as specified in 4.4.l, the argon shall contain no more than 0.02 milligram of water vapor per liter (mg/L) of gas when measured at 21.l degrees Celsius (°C) (70.0 degrees Fahrenheit (°F)) and 101.3 kilo-pascals (kPa) (l atmosphere) absolute or have a dew point of -53.9 °C (-65.0 °F) or colder.

3.3 Type I, gaseous, cylinders and valves. Unless otherwise specified, the argon shall be contained in government-owned and furnished cylinders conforming to DOT 3A or 3AA of 49 CFR 100-199 and RR-C-901. The cylinders shall be equipped with valves having outlet connection numbers 580, 677 and 680 in accordance with FED-STD-H28 and A-A-59860. When specified (see 6.2), new cylinders in accordance with RR-C-901 shall be furnished by the argon contractor and equipped with valves in accordance with A-A-59860. When specified (see 6.2) the argon shall be contained in contractor-owned DoT approved cylinders.

3.3.1 Cylinder serviceability. Government- or contractor-owned and furnished cylinders shall be inspected, maintained, and reconditioned in accordance with MIL-STD-1411 to meet the requirements of this specification and 49 CFR 100-199. The services to be performed shall be verified and deemed necessary by the applicable government quality assurance representative or a qualified agent of the purchaser prior to their accomplishment.

3.3.2 Painting. Unless otherwise specified (see 6.2), government-owned and furnished cylinders for argon shall be cleaned, treated and painted in accordance with MIL-T-704. Unless otherwise specified (see 6.2), each DoD cylinder shall be coded and marked in accordance with MIL-STD-101.

3.3.3 Filling pressures. Cylinders shall be charged to the pressure permitted (corrected to 2l.l °C) by the markings cold stamped into the shoulder of the cylinder in accordance with the CFR requirements. Filling pressure shall be tested in accordance with 4.4.5. Cylinders, except those owned by the U.S. Navy, may be filled to a pressure of 10 percent in excess of the stamped service pressure if the cylinder meets or exceeds the minimum requirements of 173.302c, 49 CFR.

3.4 Type II, liquid, containers. Type II, liquid argon, shall be contained in either contractor- or government-owned insulated containers or tank trucks as specified (see 6.2). Containers shall be in accordance with 49 CFR 171-190.

3.4.1 Container maintenance. Unless otherwise specified (see 6.2), containers that require maintenance shall be processed by the contractor for serviceability to meet the requirements of this specification and 49 CFR 171-190.

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3.5 Safety data sheet (SDS). The contracting activity shall be provided a SDS at the time of the contract award. The SDS shall be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200, Hazard Communication. When FED-STD-313 is at variance with the CFR, the CFR shall take precedence, modify, and supplement FED-STD-313. The SDS shall be included with each shipment of the material covered by this specification.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order or otherwise disapproved by the Government, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 through 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Sampling.

4.2.1 Inspection lot.

4.2.1.1 Type I argon. Every cylinder filled from the same source on the same day and offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.2.1.2 Type II argon. The entire amount of Type II argon supplied in one shipping container shall be a lot.

4.2.2 Sampling for examination of filled cylinders. A sample of filled cylinders shall be selected at random from each lot in accordance with MIL-STD-1916 at Verification Level IV, ASQ H1331 at Index Value 1.0 (associated AQLs), or other similar accept at zero (c=0) plan to verify conformance with all requirements of this specification regarding the closure, marking, and other requirements not involving testing of the argon.

4.2.3 Sampling for argon testing. Type I gaseous argon shall be sampled from each inspection lot in accordance with MIL-STD-1916 at Verification Level II, ASQ H1331 at Index Value 4.0, or other similar accept at zero (c=0) plan to verify conformance to the tests specified

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in 4.3.2. Type II liquid argon shall be sampled with a sampler, meeting the requirements of MIL-PRF-27626 or functional equivalent.

4.3 Quality conformance inspection. Quality conformance inspection shall be as follows:

4.3.1 Examination of filled cylinders. Each sample cylinder selected in accordance with 4.2.2 shall be examined for defects of the cylinder and the closure, for evidence of leakage and odor, and for unsatisfactory marking. The pressure of each Type I sample filled cylinder shall also be measured as specified in 4.4.5.

4.3.2 Argon inspection. Argon gas taken from each of the sample cylinders selected in accordance with 4.2.3 shall be subjected to the tests as specified in 4.4.1 through 4.4.4.

4.4 Argon testing.

4.4.1 Moisture impurity. Argon gas from each sample-filled cylinder shall be tested for moisture by one of the methods specified in this section after the cylinder valve, regulator and test connections have been purged by withdrawing approximately 28 liters (L) (1 cubic foot (ft^3)) of gas. In case of dispute, the accelerated gravimetric method (see 4.4.1.1 and National Bureau of Standards research paper No. 1381) shall be the standard referee test when conducted as specified herein.

4.4.1.1 Accelerated gravimetric method. The accelerated gravimetric method measures the increase in weight of phosphorous pentoxide when subjected to a flow of a measured volume of argon, as specified hereinafter. Invert the cylinder to be tested and place it in a suitable rack. Connect the cylinder to the test apparatus by the means of a length of bent, seamless, corrosion-resisting steel tubing, a high pressure union with needle valve, and a glass-to-metal joint (no rubber shall be used; a sleeve joint secured with thermoplastic is acceptable). A mercury safety valve shall be connected between the low-pressure outlet of the needle valve and the first absorber. The steel tubing shall be so fabricated that it will pass through two water baths at room temperature. The apparatus shall consist of three U-tube absorbers in series, each 10.2 centimeters (cm) (4 inches) high and 1.3 cm (1/2 inch) in diameter, containing phosphorous pentoxide on glass wool. The absorbers shall be connected through a water-saturator to a calibrated 1.4 L (0.05 ft^3) wet test meter equipped with a 1 L dial. The needle valve connecting the cylinder to the absorber shall be flushed with argon so that only gas discharged from the cylinder will pass through the absorber. The absorbers shall be brought to constant weight in a stream of dry argon before they are weighed so they will be at all times filled with argon. Before each weighing of the absorbers they shall be opened momentarily to the air, closed, and cleaned. 100 L of gaseous argon shall be discharged through the apparatus at a rate not to exceed 0.4 L per minute. The total volume of gas measured shall be corrected to conditions of 21.1 °C and 101.3 kPa pressure. The moisture content shall then be calculated from the gain in weight of the absorbents and expressed in mg/L of gas as measured at 21.1 °C and 101.3 kPa.

4.4.1.2 Freeze-out volumetric method. Moisture content may be measured by determining the amount of water collected in a trap subjected to temperatures below -183.0 °C (liquid nitrogen or oxygen) after a measured volume of argon has been passed.

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4.4.1.3 Electrical conductivity-vapor pressure method. Moisture content of argon may be determined by the electrical conductivity-vapor pressure method, as specified in the National Bureau of Standards research paper No. 1865.

4.4.1.4 Direct reading dew point method. Moisture content of argon may be measured by using direct reading dew point measurement equipment.

4.4.1.5 Other methods. Moisture content may be determined by any method regularly used by the contractor and approved by the contracting activity.

4.4.2 Oxygen impurity. The argon gas from each sample-filled cylinder shall be subjected to one of the following test procedures for determination of the oxygen impurity. If any cylinder sample of argon gas fails in any requirement, this shall be cause for rejection of the lot of filled cylinders.

4.4.2.1 Heat of reaction method. An excess of hydrogen (oxygen free) shall be mixed with the argon sample and passed over a palladium catalyst. The increase in temperature shall be measured and, by calculation, the oxygen determined for the metered flow rate of argon.

4.4.2.2 Combustion method. A small amount of oxygen-free hydrogen (previously dried) shall be mixed with the sample (previously dried). Then by combustion, the oxygen shall be combined with hydrogen to form water vapor. The amount of water shall be determined by any of the methods specified in 4.4.1. The quantity of oxygen shall be calculated from the water vapor formed and the quantity of the sample used.

4.4.2.3 Bulb method. A clear glass electric light bulb shall be modified by fusing to it, inlet and outlet connections of short lengths of glass tubing. The argon to be tested shall be passed through the bulb until a representative sample is obtained and then the gas flow shall be stopped. An 18-volt (V) electrical current shall be applied to the tungsten filament for l minute. The current shall then be turned off and the filament flashed at 85 to 100 V. Emission of smoke at the instant of flashing indicates that the argon contains more than 8 parts per million oxygen (ppm). If this test indicates presence of oxygen, it will be necessary to retest argon by another method.

4.4.2.4 Other methods. Oxygen impurity may be determined by any other method regularly used by the contractor and approved by the contracting activity.

4.4.3 Hydrogen impurity. The argon gas from each of the sample-filled cylinders shall be subjected to one of the following test methods for the determination of hydrogen impurity. If any cylinder sample of argon gas fails in any requirement, this shall be cause for rejection of the lot of filled cylinders.

4.4.3.1 Heat of reaction method. Test shall be conducted as specified in 4.4.2.1 for oxygen, except that an excess of hydrogen-free oxygen is mixed with the argon sample and passed over a palladium catalyst.

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4.4.3.2 Combustion method. Test shall be conducted as specified in 4.4.2.2 for oxygen impurity, except that small amounts of hydrogen-free oxygen (dried) shall be "burned" with a metered amount of argon sample (previously dried). The quantity of hydrogen shall be calculated from the water vapor formed and the quantity of the sample used.

4.4.3.3 Other methods. Hydrogen impurity may be determined by any other method regularly used by the contractor and approved by the contracting activity.

4.4.4 Nitrogen impurity. Argon gas from each of the sample-filled cylinders shall be subjected to one of the following test methods for determination of nitrogen impurity. If any cylinder sample of argon gas fails in any requirement, this shall be cause for rejection of the lot of filled cylinders.

4.4.4.1 Absorption in titanium metal method. Nitrogen impurity may be determined as specified in ACS Analytical Chemistry, Volume 26, No. 3.

4.4.4.2 Thermal conductivity method. Argon gas from the sample cylinder shall be passed through a thermal conductivity cell of high sensitivity where the nitrogen content is determined by comparison with argon of known nitrogen content.

4.4.4.3 Other methods. Nitrogen impurity may be determined by any alternate method regularly used by the contractor and approved by the contracting activity.

4.4.5 Filling pressure. The gauge pressure of each Type I sample cylinder shall be measured in accordance with MIL-STD-1411. Pressure-temperature conversions of CGA G-11.1 shall be used.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The argon covered in this specification is intended for use in inert gas shielded electric arc welding of aluminum, magnesium, corrosion-resisting steel, and other alloys.

6.2 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number and date of this specification.
b. Quantity and type of argon required (see 1.2).
c. When gas contractor is to furnish new gas cylinders (see 3.3).
d. When gas is to be furnished in contractor-owned cylinders (see 3.3).
e. When cylinders shall be cleaned, treated and painted in accordance with MIL-T-704 (see 3.3.2).
f. When cylinder is to be color-coded other than in accordance with MIL-STD-101 (see 3.3.2).
g. When replacement valve is other than a valve in accordance with A-A-59860 (see 3.3).
h. When liquid is to be furnished in contractor-owned containers (see 3.4).
i. Maintenance required for liquid containers (see 3.4.1).
j. Verification Level or Index Values if different from 4.2.2.2.
k. Packing requirements (see 5.1).
l. Whether cylinders should be palletized (see 5.1).
m. Whether cylinders should be marked with description of contents (see 5.1).

6.3 Purchase basis. The basis of purchase should be cubic feet of argon measured at atmospheric temperature and pressure (21.1 °C and 101.3 kPa absolute).

6.4 Cylinder contents. The contents of a full cylinder should be based on charts from the National Bureau of Standards miscellaneous publication No. 71 on the compressibility of gases, corrected for the actual volume of the cylinder filled.

6.5 Safety data sheets. Contracting officers will identify those activities requiring copies of completed safety data sheets prepared in accordance with FED-STD-313 and 29 CFR 1910.1200.

6.6 Container services. The gas contractor should furnish at no additional cost all services that are required at each filling of a container to comply with applicable regulations and normal good practice. Such services would include, but not be limited to, all inspection, testing, evacuation, and handling services required for the gas supplied.

6.7 Schedule of fees. A schedule of allowable fees should be specified by the purchaser for the gas contractor's performance of service such as the replacement of valves, valve parts, and cylinder caps, hydrostatic testing, cleaning, painting, color/coding, marking, and handling of unserviceable containers as required. All materials and components for these services should be furnished by the gas contractor.

6.8 Subject term (key word) listing.
Argon gas
Containers
Cylinders
Gaseous
Liquid

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6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy - SH
Air Force – 68
DLA - GS

Preparing activity:
DLA – GS3

Review activities:
Army – CR4
Navy – AS
Navy – MC
Navy - YD

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